



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Divya Konar
Roll No. :	20
Experiment No.:	2
Title of the Experiment:	Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis
Date of Performance:	29/07/26
Date of Submission:	29/07/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Dr. Tatwadarshi P.N.

Signature :

Date :



Aim: To apply descriptive statistical measures and data visualization techniques to summarize, analyze, and interpret the distribution and relationships present in the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

- Apply descriptive statistical and visualization techniques to explore the characteristics of a real-world healthcare dataset.
- Interpret distributions, variability, relationships, and patterns obtained from statistical analysis and visualizations.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. Matplotlib
6. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations and 8 input attributes, along with a binary outcome variable indicating diabetes status.

Attribute	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration



BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	2-Hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age in years
Outcome	1 = Diabetic, 0 = Non-diabetic

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Inspect the dataset structure, dimensions, column names, and data types.
3. Calculate descriptive statistics including mean, median, mode, minimum, maximum, variance, and standard deviation for selected numerical variables.
4. Analyze the distribution of variables using histograms and boxplots.
5. Analyze the frequency of the Outcome variable using a bar chart.
6. Generate scatter plots to study relationships such as Glucose vs. BMI and Age vs. Glucose.
7. Generate a pair plot for selected numerical variables.
8. Interpret the statistical summaries and visualizations and record important observations.

Description of the Experiment

Descriptive statistics summarize the important characteristics of a dataset using numerical measures, while data visualization presents these characteristics graphically.

The experiment follows the workflow: Dataset, Descriptive Statistics, Distribution Analysis, Visualization, Relationship Analysis, Interpretation,

Students will use the Pima Diabetes Dataset to analyze the central tendency, variability, distributions, and relationships among health-related variables.



Detailed Description of the Statistical techniques:

Mean, Median, and Mode

These measures describe the central or typical value of a dataset.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The median is the middle value, while the mode is the most frequently occurring value.

Range, Variance, and Standard Deviation

These measures describe data variability.

$$Range = Maximum - Minimums^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} s = \sqrt{s^2}$$

Histogram

Used to study the distribution, spread, and shape of numerical variables such as Glucose, BMI, and Age.

Boxplot

Used to visualize the median, quartiles, spread, and potential outliers.

$$IQR = Q_3 - Q_1$$

Bar Chart

Used to display the frequency of categorical or binary variables such as Outcome.

Scatter Plot

Used to examine the relationship between two numerical variables, such as:

- Glucose and BMI
- Age and Glucose



Pair Plot

Used to simultaneously visualize pairwise relationships and distributions among multiple numerical variables.

Conclusion

Descriptive statistical measures and visualization techniques were applied to the Pima Indians Diabetes Dataset to understand the distribution, variability, and relationships among the variables.

Output :

```
[Running] python -u "e:\MYDATA\STATS\IMAGES\exp2.py"
===== FIRST 5 ROWS =====
| Pregnancies  Glucose  BloodPressure  ...  DiabetesPedigreeFunction  Age  Outcome
0            6     148             72  ...                0.627    50      1
1            1      85             66  ...                0.351    31      0
2            8     183             64  ...                0.672    32      1
3            1      89             66  ...                0.167    21      0
4            0     137             40  ...                2.288    33      1

[5 rows x 9 columns]

===== DATASET SHAPE =====
(768, 9)

===== COLUMN NAMES =====
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='str')

===== DATA TYPES =====
Pregnancies      int64
Glucose           int64
BloodPressure     int64
SkinThickness     int64
Insulin           int64
BMI               float64
DiabetesPedigreeFunction float64
Age               int64
Outcome           int64
dtype: object
```



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===== DATASET INFO =====

<class 'pandas.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 54.1 KB

===== MISSING VALUES =====

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

Age	29.0000
Outcome	0.0000

dtype: float64

Mode	
Pregnancies	1.000
Glucose	99.000
BloodPressure	70.000
SkinThickness	0.000
Insulin	0.000
BMI	32.000
DiabetesPedigreeFunction	0.254
Age	22.000
Outcome	0.000

Name: 0, dtype: float64

Minimum	
Pregnancies	0.000
Glucose	0.000
BloodPressure	0.000
SkinThickness	0.000
Insulin	0.000
BMI	0.000
DiabetesPedigreeFunction	0.078
Age	21.000
Outcome	0.000

dtype: float64

Maximum	
Pregnancies	17.00
Glucose	199.00
BloodPressure	122.00

===== DESCRIPTIVE STATISTICS =====

	Pregnancies	Glucose	...	Age	Outcome
count	768.000000	768.000000	...	768.000000	768.000000
mean	3.845052	120.894531	...	33.240885	0.348958
std	3.369578	31.972618	...	11.760232	0.476951
min	0.000000	0.000000	...	21.000000	0.000000
25%	1.000000	99.000000	...	24.000000	0.000000
50%	3.000000	117.000000	...	29.000000	0.000000
75%	6.000000	140.250000	...	41.000000	1.000000
max	17.000000	199.000000	...	81.000000	1.000000

[8 rows x 9 columns]

Mean	
Pregnancies	3.845052
Glucose	120.894531
BloodPressure	69.105469
SkinThickness	20.536458
Insulin	79.799479
BMI	31.992578
DiabetesPedigreeFunction	0.471876
Age	33.240885
Outcome	0.348958

dtype: float64

Median	
Pregnancies	3.0000
Glucose	117.0000
BloodPressure	72.0000
SkinThickness	23.0000
Insulin	30.5000
BMI	32.0000
DiabetesPedigreeFunction	0.3725

BloodPressure	122.00
SkinThickness	99.00
Insulin	846.00
BMI	67.10
DiabetesPedigreeFunction	2.42
Age	81.00
Outcome	1.00

dtype: float64

Variance	
Pregnancies	11.354056
Glucose	1022.248314
BloodPressure	374.647271
SkinThickness	254.473245
Insulin	13281.180078
BMI	62.159984
DiabetesPedigreeFunction	0.109779
Age	138.303046
Outcome	0.227483

dtype: float64

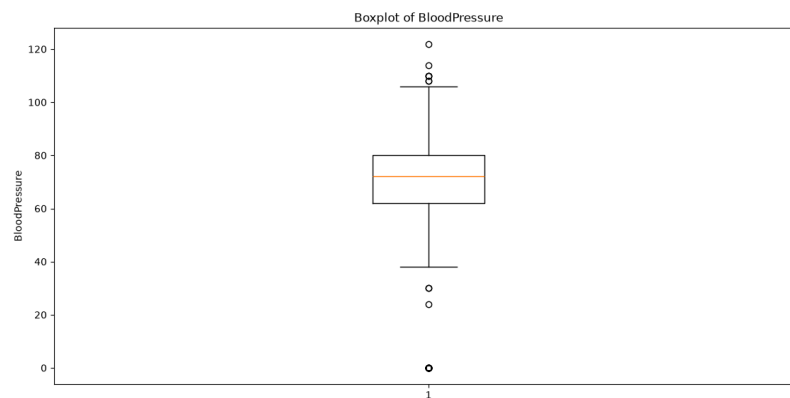
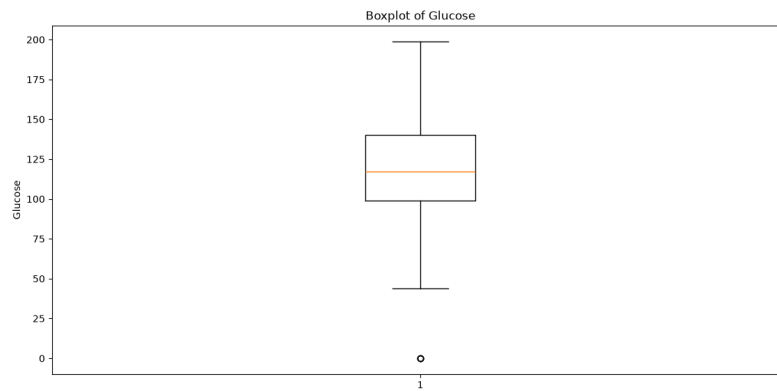
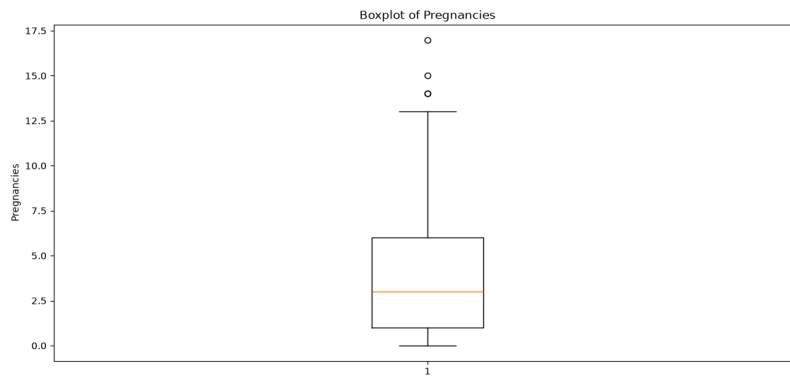
Standard Deviation	
Pregnancies	3.369578
Glucose	31.972618
BloodPressure	19.355807
SkinThickness	15.952218
Insulin	115.244002
BMI	7.884160
DiabetesPedigreeFunction	0.331329
Age	11.760232
Outcome	0.476951

dtype: float64



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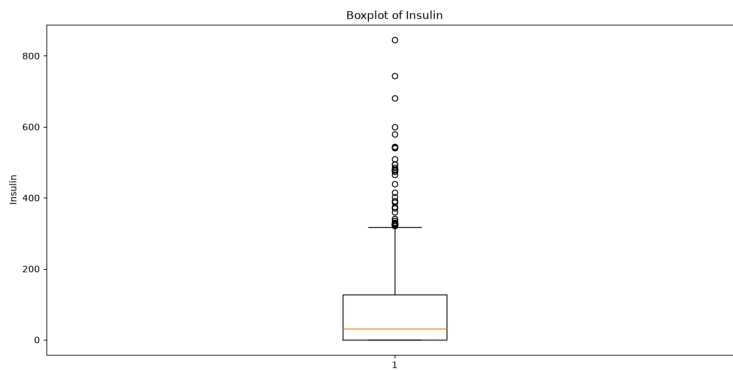
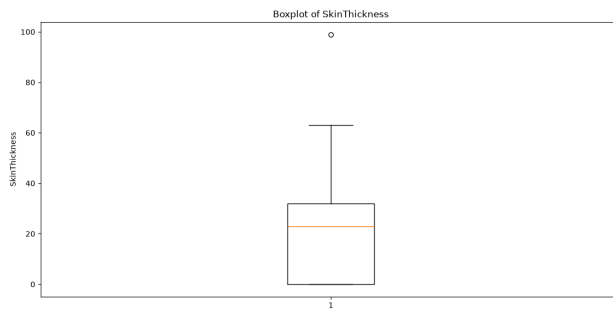
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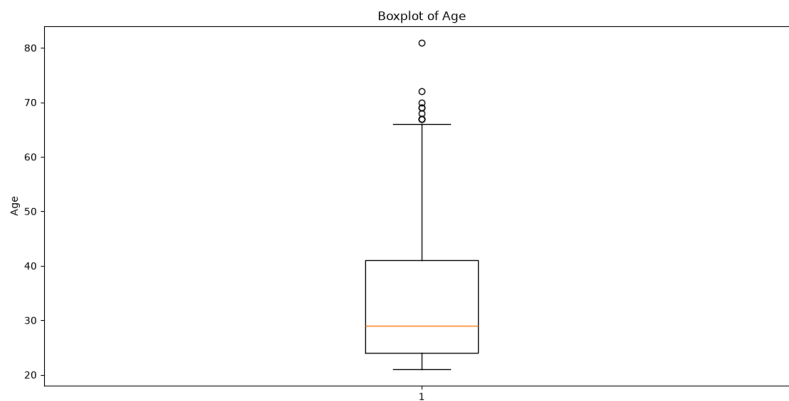
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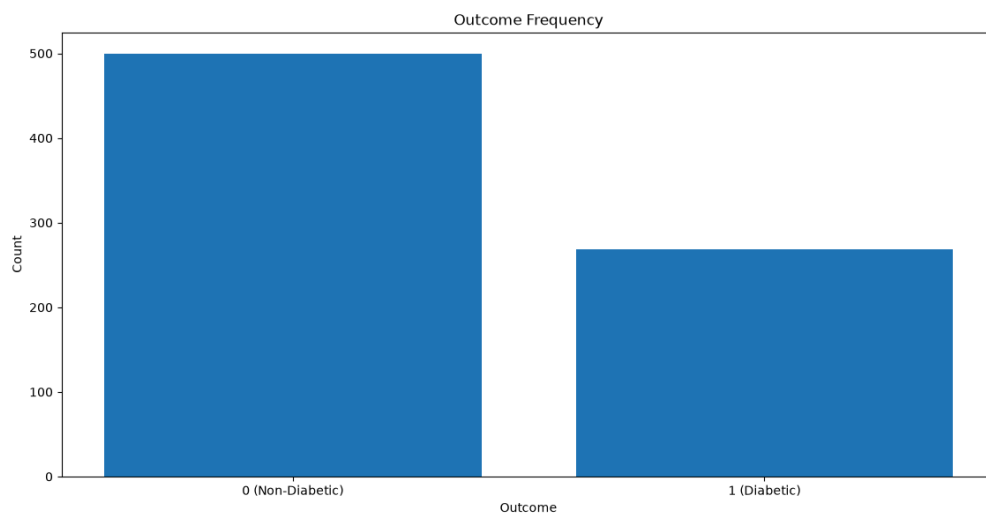
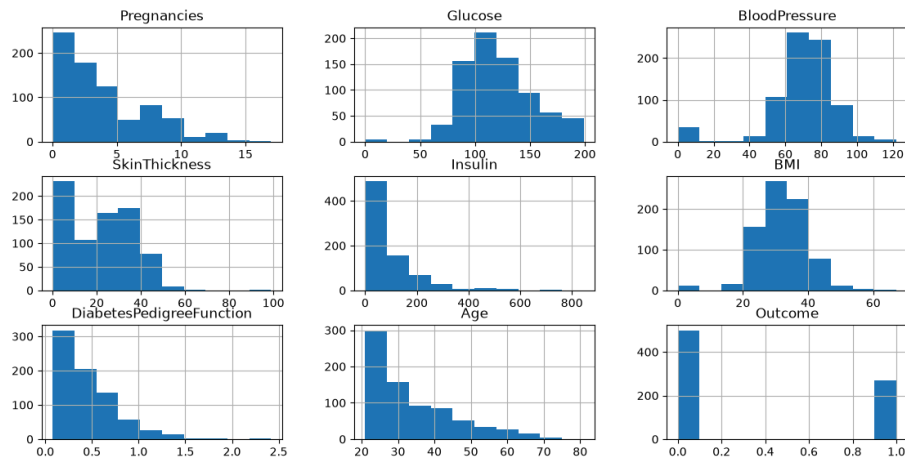


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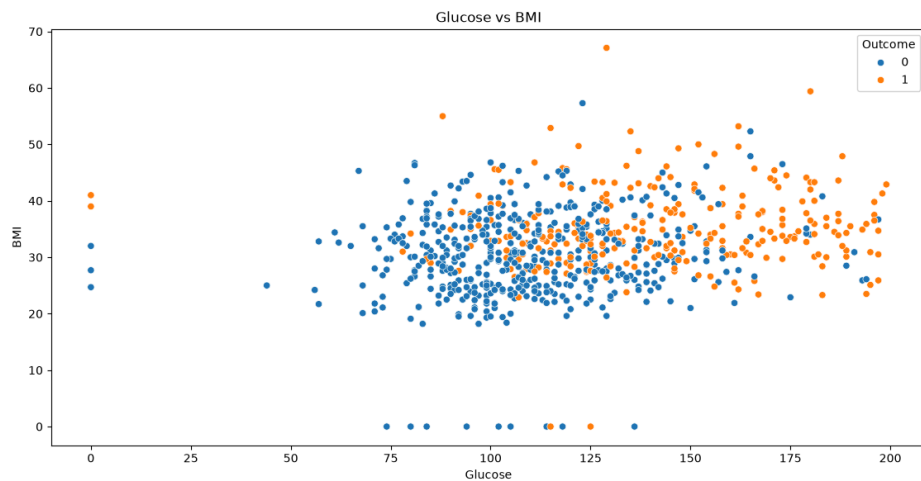
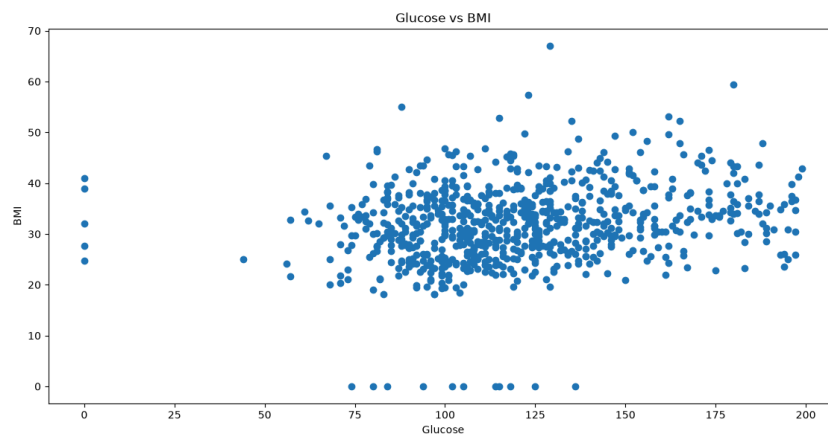
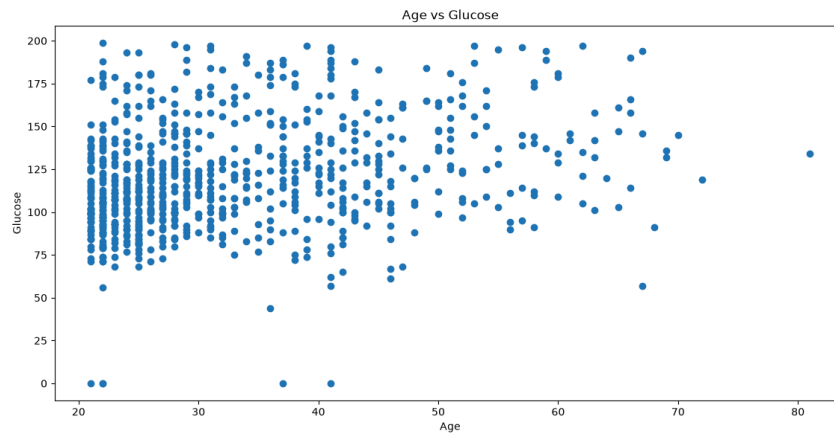
Histogram of Numerical Variables





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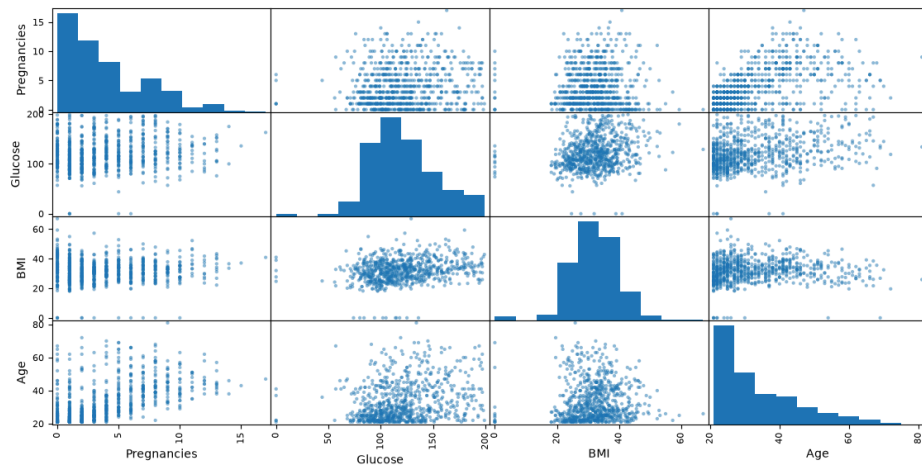




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Pair Plot





Questions to be Answered by Students

1. Which numerical variable has the highest standard deviation?

ANS : Insulin

2. Which variable shows the greatest difference between its mean and median?

ANS : Insulin

3. What relationship can be observed between Glucose and BMI from the scatter plot?

ANS : A weak positive relationship (higher BMI tends to have higher glucose)

4. State two important insights obtained from the analysis.

ANS : 1. Higher glucose is associated with diabetes

2. BMI and Age tend to be higher in diabetic patients

5. Which variable(s), if any, contain potential outliers? How were these identified?

ANS : Insulin , Skin thickness , BMI , DiabetesPedigreeFunction , and Pregnancies. They were identified using box plots (IQR) method.

6. State **two important insights** obtained from your exploratory statistical analysis of the dataset.

ANS : 1. Several variables contains outliers and invalid zero values.

2. Glucose is the strongest indicator of diabetes among the numerical values.



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